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Effect of deep trapping states on space charge suppression in polyethylene/ZnO nanocomposite

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This letter intends to reveal the mechanism of space charge suppression in low density polyethylene (LDPE)/ZnO nanocomposites. Trap level and space charge distributions were obtained from modified isothermal discharge current method and pulsed electro-acoustic (PEA) method, respectively. The results showed that ZnO nanoparticle doping introduced large amounts of deep trapping states, significantly reduced space charge accumulation and conduction current. The results can be explained in terms of deep trapping states resulted from the interface regions and morphology structure changes by nanoparticles doping, which greatly reduced the charge mobility, raised the charge injection potential at the contact and weakened impurity ionization. © 2011 American Institute of Physics. [doi:10.1063/1.3646909]

Polymer electrical insulation materials are apt to capture charge carriers to form space charges under high electrical field, which is an especially major restricted factor for high voltage direct current (HVDC) cable development.¹ Space charges induce kinds of detrimental effects, seriously distorting the electrical field, generating hot electrons, leading to electrical-mechanical energy storage and release, etc.² Thus, the investigation of space charge suppression method and mechanism has become a main focus in the development of HVDC cable.

Recently, nanoparticle fillers are widely used for enhancing electrical properties of polymer insulation materials, especially space charge suppression under high direct current (DC) electrical field.^{3,4} This letter intends to reveal the microscopic mechanism of space charge suppression in low density polyethylene (LDPE)/ZnO nanocomposite via deep trapping states determination and conduction current measurement.

The nanocomposites were prepared by melt blending method. The ZnO nanoparticles have an average particle size of about 30 nm and were surface treated with 2 wt. % silane coupling agent 3-Aminopropyltriethoxysilane. LDPE and surface treated ZnO nanoparticles were melt-blended in a rheometer at 423 K. The obtained nanocomposites were pressing molded at 313 K under a pressure of 10 MPa to obtain films with a thickness of about 200 μm . All samples were evaporated with Al electrode on both sides with a diameter of 25 mm.

The sample section surface, fractured in liquid nitrogen, was observed by scanning electron microscopy (SEM, Philips Quanta 200) operated at 20 kV in order to observe the dispersion of nanofillers. The trap level distribution was measured based on the modified isothermal discharge current method, which has been described in detail in previous pub-

lications by the author.⁵ First, charges were injected into the sample under 30 kV/mm at 323 K for 40 min. Then, the sample was short-circuited and the discharge current was measured at 323 K. Conduction current was measured by a picoammeter EST122 and a data acquisition system under DC 10 kV/mm and 50 kV/mm, respectively. Space charge distribution measurement was performed based on pulsed electro-acoustic (PEA) method under DC 50 kV/mm at room temperature. The principle and application of PEA method have been widely discussed in previous publications.⁶

Dispersion of the ZnO nanoparticles in LDPE matrix is presented by SEM image in Figure 1. The ZnO nanoparticles are uniformly distributed in LDPE. The particle size is around 30~80 nm. No large agglomerate can be observed. The border between the nanoparticles and polymer matrix is obscure, indicating significant interface region.

Figure 2 shows the trap level distributions of LDPE and LDPE/ZnO nanocomposite. The results show that the trap levels are both in the range 0.8~1.0 eV with a peak density at 0.94 eV, which can be regarded as deep trap levels in LDPE.^{7,8} The peak trap level density in LDPE and LDPE/ZnO nanocomposite are 8×10^{20} and $4.8 \times 10^{21}/(\text{eV} \cdot \text{m}^3)$,

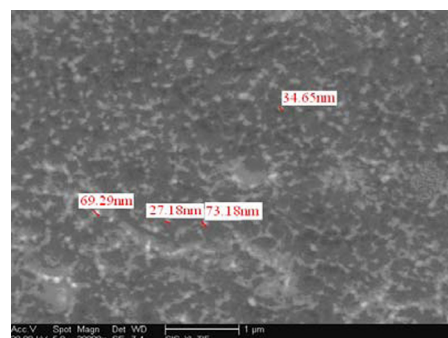


FIG. 1. (Color online) SEM section images of LDPE/ZnO 7% treated.

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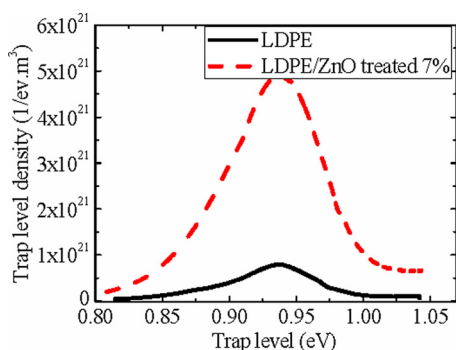


FIG. 2. (Color online) Trap level distribution LDPE and LDPE/ZnO 7% treated.

respectively, which are in the same range as the existing results ($10^{16} \sim 10^{22} \text{ m}^{-3} \cdot \text{eV}$).^{5,8} As can be seen clearly, large amounts of deep traps are introduced by ZnO nanoparticle doping. Note that the space charge accumulation in the bulk of the nanocomposite is significantly suppressed (as shown later by PEA measurement), so the measured trap states are mainly located close to the surface.

LDPE is a semi-crystalline polymeric material with most of the deep trapping states (about 0.92 eV proposed by Ieda) formed by physical defects mainly located in the interface regions, which most strongly affects the carrier transport in LDPE.⁸ The structure of the interface regions involving physical defects, such as molecular entanglement and kinks, chain loops, chain ends, and crosslinks between the amorphous and crystalline regions, is likely to create cavity traps, which is now widely accepted in polyethylene (PE). The radius of cavity traps is about $\sim 10 \text{ nm}$ for coulombic traps and 0.1 nm for neutral traps.⁹ The depth of cavity trap depends upon the size of cavity and it is approximately proportional to R^{-2} (R is the cavity radius). So, it is reasonable that the cavity trap in the interface regions with molecules tightly bounded by crystallites is deeper than in the amorphous regions since the size of cavity is smaller for the former.¹⁰

Nanoparticle doping can generate enormous interface areas due to their huge specific surface area. The local physical properties of the interface polymers are different in free volume, chain arrangement, mobility, etc. This may increase the density and depth of the cavity trapping states.^{11–13} In addition, the nanoparticles can act as heterogeneous nucleation agents during the polymer crystallization process. This can lead to the disruption of the large spherulites and result in an increase in the interface areas. As stated before, cavity trapping states are readily generated in these areas. Since the trap levels in the nanocomposite are in the same range as in pure LDPE, so the origin of the traps should be cavities rather than chemical impurity states introduced by nanoparticle surface treatment in nature as described in LDPE.

Figure 3 describes the conduction current measurement of LDPE and LDPE/ZnO. The conduction current of the nanocomposite is about 1/10 and 1/3 of that in LDPE under 10 kV/mm and 50 kV/mm, respectively. Both the low field and high field conduction current are correlated to the charge mobility.¹⁴ The measured charge mobility for LDPE is very low from 10^{-15} to $10^{-7} \text{ m}^2/\text{V} \cdot \text{s}$ and it shows a thermally activated process via adjacent trapping states. As has been

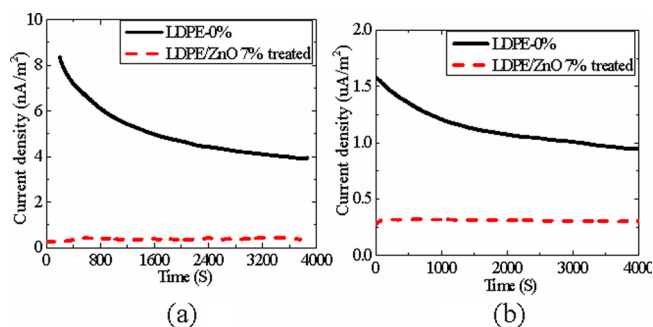


FIG. 3. (Color online) Conduction current of LDPE and LDPE/ZnO 7% treated under 10 kV/mm and 50 kV/mm.

showed clearly in Figure 2, the trap level density in the nanocomposite is about six times of that in LDPE. So, the conduction current reduction in the nanocomposite should be a result of the trap density increase. This, in turn, confirmed that the deep traps are introduced by ZnO nanoparticle doping.

The space charge distribution is shown in Figure 4. Immediately after the voltage is applied, both electrons and holes are injected from the cathode and anode separately and symmetrically in LDPE. After 900 s, large amounts of heterocharges and homocharges are formed around the cathode, with a maximum density of about 6 C/m^3 . Heterocharges come from the ionization of impurities in LDPE. Positive ions are saturated and accumulated around the cathode quickly since it is difficult for them to neutralize via mass transport across the electrode interface. As heterocharges are formed near the cathode, the electric field near the cathode is enhanced significantly, which decreases the potential barrier for electron injection. So, the injected holes are generally covered.

However, what calls special attention is that hardly any detectable homocharges or heterocharges are formed in the LDPE/ZnO nanocomposite. This reveals that LDPE/ZnO nanocomposite exhibits remarkable space charge suppression ability, which is of great significance in practice for HVDC cable insulation development.

Tanaka¹¹ have attributed the reduced space charge accumulation in nanocomposite to the possible introduction of shallow trapping states and the resulting enhancement in charge mobility.¹¹ Obviously, this model is inconsistent with our results, which shows significant increase in the deep trap level densities and decrease in charge mobility in the LDPE/ZnO nanocomposite.

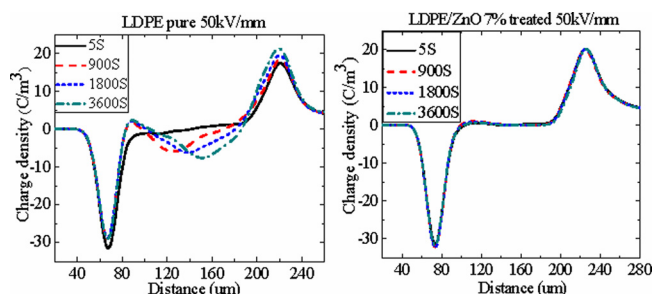


FIG. 4. (Color online) Space charge distribution in LDPE and LDPE/ZnO 7% treated under 50 kV/mm.

In order to reveal the space charge suppression mechanism in LDPE/ZnO nanocomposite, the author focuses on the effect of the large amounts of deep trapping states in the nanocomposite. When electrons (or holes) are injected from the cathode (or anode), they are readily captured by the deep trapping states in the interface between the polymer and the electrode, where they are too close to the electrode to be resolved by current space charge detection methods. The trapped homocharges form a charge injection blocking by decrease the efficient electric field at the interface and raise the potential barrier for carrier injection. In addition, positive ions and electrons due to impurity ionization are limited to a strongly localized region as a result of the low charge mobility, which weakened the ionization. Thus, hetero-space charges are restricted.

In conclusion, it has been demonstrated that ZnO nanoparticles doping can remarkably suppress space charge accumulation and significantly reduce conductivity under low and high electric field in LDPE. This has been attributed to the introduction of deep trapping states resulted from the generated interface regions by nanometer ZnO doping, which can largely reduce charge carrier mobility. The investigation has its potentials for development of HVDC cable insulations.

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- ¹T. L. Hanley, R. P. Burford, R. J. Fleming, and K. W. Barber, *IEEE Electr. Insul. Mag.* **19**(1), 13 (2003).
- ²G. Mazzanti, G. C. Montanari, and L. A. Dissado, *IEEE Trans. Dielectr. Electr. Insul.* **12**(5), 876 (2005).
- ³T. Takada, Y. Hayase, Y. Tanaka, and T. Okamoto, *IEEE Trans. Dielectr. Electr. Insul.* **15**(1), 152 (2008).
- ⁴X. Huang, P. Jiang, and Y. Yin, *Appl. Phys. Lett.* **95**, 242905 (2009).
- ⁵L. Qingquan, T. Fuqiang, Y. Chun, H. Lijuan, and W. Yi, *J. Electrostat.* **68**(3), 243 (2010).
- ⁶T. Maeno and K. Fukunaga, *IEEE Trans. Dielectr. Electr. Insul.* **3**(6), 754 (1996).
- ⁷X. Wang, N. Yoshimura, Y. Tanaka, K. Murata, and T. Takada, *J. Phys. D: Appl. Phys.* **31**(16), 2057 (1998).
- ⁸M. Ieda, *IEEE Trans. Electr. Insul.* **19**(3), 162 (1984).
- ⁹T. Mizutani, Y. Suzuoki, M. Hanai, and M. Ieda, *Jpn. J. Appl. Phys.* **21**(11), 1639 (1982).
- ¹⁰M. Ieda, T. Mizutani, and Y. Suzuoki, *Memoirs of the Faculty of Engineering* (Nagoya University, Japan, 1980) Vol. **32**, p. 173.
- ¹¹T. Tanaka, *IEEE Trans. Dielectr. Electr. Insul.* **12**(5), 914 (2005).
- ¹²F. Tian, W. Bu, L. Shi, C. Yang, Y. Wang, and Q. Lei, *J. Electrostat.* **69**(1), 7 (2010).
- ¹³J. K. Nelson, *Dielectric Polymer Nanocomposites* (Springer, New York, 2010).
- ¹⁴J. A. Anta, G. Marcelli, M. Meunier, and N. Quirke, *J. Appl. Phys.* **92**(2), 1002 (2002).